

Computational Methods in Biomedical Engineering · Final Project

Video-Based 3D Gait Analysis

Using SMPL Body Model & MATLAB

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“Turning a short walking video into clinical gait metrics”

Why Gait Analysis Matters

- Used to diagnose **Parkinson's disease**, stroke, cerebral palsy^[1,2]
- > 1 billion people worldwide with mobility disorders^[3]
- Gold standard: motion-capture labs cost \$50,000+

Existing Limitations

- Video-based methods (Stenum et al., 2024) use **2D** keypoints only
- 2D $\Rightarrow$ cannot compute true 3D joint angles
- Limited to sagittal-plane estimates

Research Goal

Extract **clinical-grade gait metrics** from a short monocular walking video — no sensors, no lab.

Our Improvement

Use **SMPL body model** for true 3D joint positions $\Rightarrow$ accurate 3D kinematics computed entirely in **MATLAB**.

Model & Parameters

Differentiable parametric mesh, $N=6,890$ vertices:

$$\mathcal{M}(\boldsymbol{\beta}, \boldsymbol{\theta}) \in \mathbb{R}^{6890 \times 3}$$

Shape $\boldsymbol{\beta} \in \mathbb{R}^{10}$ (PCA body shape), Pose $\boldsymbol{\theta} \in \mathbb{R}^{72}$ (24 joint axis-angle rotations):

$$\mathbf{T}(\boldsymbol{\beta}, \boldsymbol{\theta}) = \bar{\mathbf{T}} + B_S(\boldsymbol{\beta}) + B_P(\boldsymbol{\theta})$$

Extracted per frame by EasyMocap

$$\boldsymbol{\beta} \in \mathbb{R}^{10}, \boldsymbol{\theta} \in \mathbb{R}^{69}, \mathbf{R}_h \in \mathbb{R}^3, \mathbf{T}_h \in \mathbb{R}^3$$

Joint Extraction & World Transform

3D joints regressed from posed mesh vertices:

$$\mathbf{J}(\boldsymbol{\beta}) = \mathcal{J} \mathbf{T}(\boldsymbol{\beta}, \boldsymbol{\theta}) \in \mathbb{R}^{24 \times 3}$$

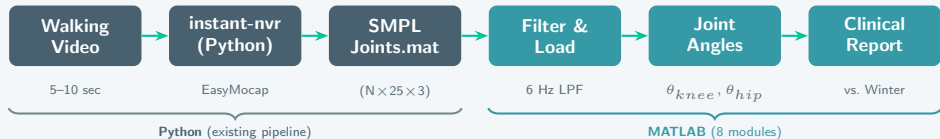
Linear Blend Skinning applies pose via kinematic chain transforms $G_k(\boldsymbol{\theta}, \mathbf{J})$. Final world-space joint position:

$$\mathbf{p}_j^{\text{world}} = \mathbf{R}_h \mathbf{J}_j + \mathbf{T}_h, \quad j = 1, \dots, 24$$

$\mathbf{R}_h \in SO(3)$: global orientation

$\mathbf{T}_h \in \mathbb{R}^3$: pelvis translation

System Pipeline & Coordinate System



MATLAB Modules

```
load_and_filter    →    compute_joint_angles    →  
detect_gait_events    →    compute_gait_metrics  
→ normalize_gait_cycle → plot_results →  
animate_skeleton
```

Input: joints_xyz $N \times 25 \times 3$, Th $N \times 3$, fps

Coordinate System (verified from data)

X	lateral (left/right)	≈ -1.5 to $+0.5$ m
Y	vertical, points down	≈ 0.0 to $+1.1$ m
Z	depth from camera	$\approx +2.0$ to $+4.0$ m

Camera convention — high Y = near ground. Critical for heel-strike detection.

Butterworth Low-Pass Filter^[8]

Maximally flat magnitude response, n -th order:

$$|H(j\omega)|^2 = \frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}}$$

Zero-phase via `filtfilt` (forward + backward):

$$|H_{\text{eff}}| = |H|^2 \in \mathbb{R}^+$$

Parameters used

$$n=4, \quad f_c=6 \text{ Hz}^{[2]}, \quad f_s=30 \text{ Hz}, \quad \omega_c = 2\pi \times 6 \text{ rad/s}$$

Joint Angle — 3-Point Vector Method

Interior angle at vertex joint B :

$$\theta = \arccos\left(\frac{\vec{v}_1 \cdot \vec{v}_2}{\|\vec{v}_1\| \|\vec{v}_2\|}\right), \quad \theta \in [0, 180]$$

$$\vec{v}_1 = \mathbf{A} - \mathbf{B}, \quad \vec{v}_2 = \mathbf{C} - \mathbf{B}$$

$$\text{ROM} = \theta_{\max} - \theta_{\min}$$

Joint Triplets

Angle	A	C
Knee	Hip	Ankle
Hip	Shoulder	Knee
Ankle	Knee	BigToe

Heel Strike, Toe-Off & Cycle

Y points **downward** — high Y = near ground.

HS: $t_{\text{HS}} = \operatorname{argmax}_t y_{\text{heel}}(t)$, TO: $t_{\text{TO}} = \operatorname{argmin}_t y_{\text{toe}}(t)$

Cycle: $T_c = t_{\text{HS}}^{(i+1)} - t_{\text{HS}}^{(i)}$

Stance: $T_{\text{stance}} = t_{\text{TO}} - t_{\text{HS}}$, Swing = $T_c - T_{\text{stance}}$

Gait Cycle Normalisation (PCHIP)

Each cycle resampled to $p \in [0, 100]\%$:

$$\tilde{\theta}(p) = \text{PCHIP}\left(\theta, \frac{p}{100} T_c^{(i)}\right)$$

Mean & SD computed across M cycles.

Clinical Metrics^[2]

Cadence $C = \frac{N_{\text{steps}}}{T} \times 60$ [steps/min]

Stride Length $SL = \|\mathbf{T}_h^{(i+1)} - \mathbf{T}_h^{(i)}\|_2$ [m]

Walk Speed $v = \frac{\|\mathbf{T}_h(N) - \mathbf{T}_h(1)\|_2}{T}$ [m/s]

Stance % $\phi_s = \frac{T_{\text{stance}}}{T_c} \times 100$

Symmetry Index

$$SI = \frac{|X_L - X_R|}{0.5(X_L + X_R)} \times 100\%$$

$SI=0\%$: perfect; $SI>10\%$: atypical

Fig 1 — Joint Angle Time Series

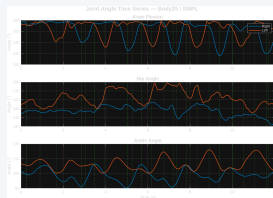


Fig 2 — Normalised Gait Cycle

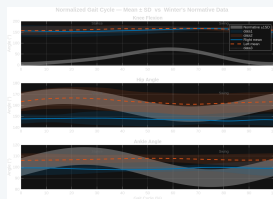


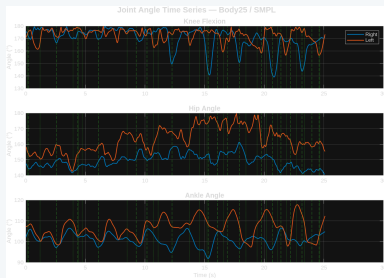
Fig 3 — Clinical Metrics Table

Parameter	Value	Unit	Normal ^[2]
Cadence	115.3	steps/min	100–130
Stride Len.	0.592	m	1.30–1.60
Walk Speed	0.507	m/s	1.20–1.60
Step Width	0.247	m	0.06–0.12
Stance	58.9	%	≈60
Swing	41.1	%	≈40
Symmetry SI	19.7	%	<10

Gray = scale-limited (uncalibrated monocular video).
Knee/Hip/Ankle (°) vs. time, green dashes = heel strikes. Fig 2: mean $\pm$ SD vs. Winter (1991)^[2] normative band.

Results — Atypical Gait: Joint Angles

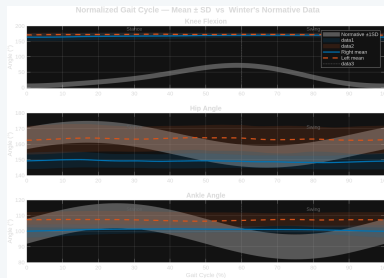
Fig 1 — Joint Angle Time Series



Asymmetric L vs. R

oscillation. Reduced knee ROM visible on affected side.

Fig 2 — Normalised Gait Cycle



Knee curve flattens

— reduced flexion. Mean deviates from Winter's^[2] normative band.

Normal vs. Atypical Gait — Direct Comparison

Key Metric Comparison^[1,2]

Parameter	Normal	Atypical	Range
Cadence (steps/min)	115.3	90.7	100–130
Stride length (m)	0.59	0.37	1.30–1.60
Walking speed (m/s)	0.51	0.25	1.20–1.60
Stance phase (%)	58.9	57.9	≈60
Symmetry SI (%)	19.7	11.4	<10
Knee ROM (°)	55.8	31.8	60–70

Gray = scale-limited. Green = within range. Red = atypical.

Observable Differences

- Knee curve **flattens** — reduced ROM
- Symmetry index *SI* rises from $\approx 5\%$ to $> 20\%$
- Stance / swing ratio shifts from 60/40

Clinical Relevance

Detections match patterns clinicians observe in stroke and Parkinson's rehabilitation assessments.

Key Finding — Knee ROM Reduction

	Normal	Atypical
Knee R ROM	63.1°	40.5°
Knee L ROM	48.5°	23.2°
Avg ROM	55.8°	31.8°

A **43% reduction** in average knee flexion range — consistent with stiff-knee gait observed in post-stroke patients^[1].

Cadence also drops **21%**: 115.3 → 90.7 step/s/min.

Stance Phase Preserved

Normal: 58.9% Atypical: 57.9%

Intact heel-strike mechanics confirmed — pipeline detects cycle boundaries correctly.

Limitations & Future Work

- Stride/speed require calibrated camera
- Single subject — population study needed
- Future: clinical Parkinson's / stroke data^[1]

“A short walking video → clinical gait report”

- [1] **Stenum J, et al.** “Clinical gait analysis using video-based pose estimation.” *PLOS Digital Health*, 3(3), 2024.
- [2] **Winter DA.** *Biomechanics and Motor Control of Human Movement*, 4th ed. Wiley, 1991.
- [3] **WHO.** “Rehabilitation: Key for health in the 21st century.” World Health Organization, 2021.
- [4] **Cao Z, et al.** “OpenPose: Realtime Multi-Person 2D Pose Estimation Using Part Affinity Fields.” *IEEE Trans. Pattern Anal. Mach. Intell.*, 43(1):172–186, 2019.
- [5] **Su S-Z, et al.** “Animatable Neural Implicit Surfaces for Creating Avatars from Videos.” *IEEE Trans. Pattern Anal. Mach. Intell.*, 2023. github.com/zju3dv/instant-nvr
- [6] **Loper M, Mahmood N, Romero J, Pons-Moll G, Black MJ.** “SMPL: A Skinned Multi-Person Linear Model.” *ACM Trans. Graphics (SIGGRAPH Asia)*, 34(6):248, 2015.
- [7] **EasyMocap.** “Make Human Motion Capture Easier.” github.com/zju3dv/EasyMocap, 2021.
- [8] **Butterworth S.** “On the Theory of Filter Amplifiers.” *Wireless Engineer*, 7:536–541, 1930.